

# Clinical Decision-Support: Comprehensive Genetic & Dermatological Dossier

PHYSICIAN & MEDICAL GENETICS EDITION | Record ID: MM-CLAIRE-389B | Model: Hybrid-XGB-AdaBoost-VQC-Derm-v2.0

|                      |                                                                       |                          |                                            |
|----------------------|-----------------------------------------------------------------------|--------------------------|--------------------------------------------|
| Patient Demographics | Age: 44.0 yrs   Biological Sex: Female<br>Detected Phototype: Type II | Tri-Model Consensus Risk | 25% (MODERATE RISK)                        |
| Dual Boosting + VQC  | XGBoost (0.45) + AdaBoost (0.25) + 4-Qubit VQC (0.30)                 | Epistemic Uncertainty    | Confidence: High ( $\sigma = \pm 0.1333$ ) |

## 1. Detailed Molecular & Genetic Deficiencies Analysis

| Genetic Pathway / Defect                     | Status / Score    | Classification | Clinical Oncologic Implications                                                                                     |
|----------------------------------------------|-------------------|----------------|---------------------------------------------------------------------------------------------------------------------|
| TP53 Tumor Suppressor (17p13.1 DNA Binding)  | Score: 0.15 / 1.0 | Wild-Type      | Loss of apoptotic checkpoint control, impaired p21/CDKN1A induction, elevated genomic destabilization.              |
| BRCA1 / BRCA2 (HR Repair Pathway)            | Absent (0.0)      | Intact         | Homologous recombination double-strand break repair failure; high sensitivity to PARP inhibition & platinum agents. |
| Tumor Mutational Burden (Somatic Density)    | 3.1 mut/Mb        | Low / Moderate | Elevated somatic neoantigen expression. Biomarker for immune checkpoint inhibitor response (Anti-PD-1/PD-L1).       |
| hs-CRP / Inflammation (Microenvironment)     | 1.6 mg/L          | Normal         | Sustained pro-inflammatory cytokines (IL-6 / TNF-alpha) facilitating cutaneous angiogenesis & immune evasion.       |
| Hereditary Pedigree (First-Degree Relatives) | Confirmed (1.0)   | POSITIVE       | Germline susceptibility inheritance profile. Multi-generational screening indicated.                                |

## 2. Dual Boosting (XGBoost + AdaBoost) & Parameterized Quantum VQC Ensemble

| Inference Engine                     | Estimated Probability | Weight | Algorithm Architecture & Role                                                                                   |
|--------------------------------------|-----------------------|--------|-----------------------------------------------------------------------------------------------------------------|
| XGBoost (Gradient Boosted Trees)     | 31%                   | 45%    | Depth-6 gradient boosting over combined 17-dimensional cutaneous & genomic vector.                              |
| AdaBoost (Adaptive Stumps)           | 64%                   | 25%    | Sequential decision stumps emphasizing borderline cases with ambiguous melanin-genomic interplay.               |
| Quantum VQC (Qiskit Aer / PennyLane) | 13%                   | 30%    | 4-qubit parameterized variational circuit with angle embedding exploring non-linear Hilbert space correlations. |
| Hybrid Calibrated Consensus          | 25%                   | 100%   | Calibrated consensus with discordance variance boundary ( $\pm 0.208$ ). Stratified to HIGH RISK.               |

## 3. Model Risk Spectrum Across All Skin Phototypes (Fitzpatrick Types I–VI)

Simulates the patient's exact genetic mutations across all skin pigmentation categories to evaluate cutaneous susceptibility:

| Fitzpatrick Phototype                   | Optical ITA / Melanin  | XGBoost | AdaBoost | Hybrid Consensus | Stratification Tier |
|-----------------------------------------|------------------------|---------|----------|------------------|---------------------|
| Type I (Very Light) [PATIENT PHENOTYPE] | ITA: 78.0°   Mel: 3.5  | 49%     | 42%      | 45%              | Moderate Risk       |
| Type II (Light) [PATIENT PHENOTYPE]     | ITA: 52.0°   Mel: 14.0 | 47%     | 41%      | 44%              | Moderate Risk       |

|                                   |                         |     |     |     |               |
|-----------------------------------|-------------------------|-----|-----|-----|---------------|
| Type III (Intermediate)           | ITA: 35.0°   Mel: 22.0  | 45% | 39% | 42% | Moderate Risk |
| Type IV (Tan / Olive)             | ITA: 18.0°   Mel: 30.0  | 44% | 38% | 41% | Moderate Risk |
| Type V (Brown)                    | ITA: -12.0°   Mel: 46.0 | 41% | 36% | 39% | Moderate Risk |
| Type VI (Dark / Deeply Pigmented) | ITA: -62.0°   Mel: 72.0 | 36% | 32% | 34% | Moderate Risk |

4. Actionable Clinical Guidance & Triage Protocol

Routine preventive maintenance

**MANDATORY MEDICAL DISCLAIMER:** AI-generated decision-support output. Not a final medical diagnosis. Final clinical decision remains with a qualified healthcare professional.